

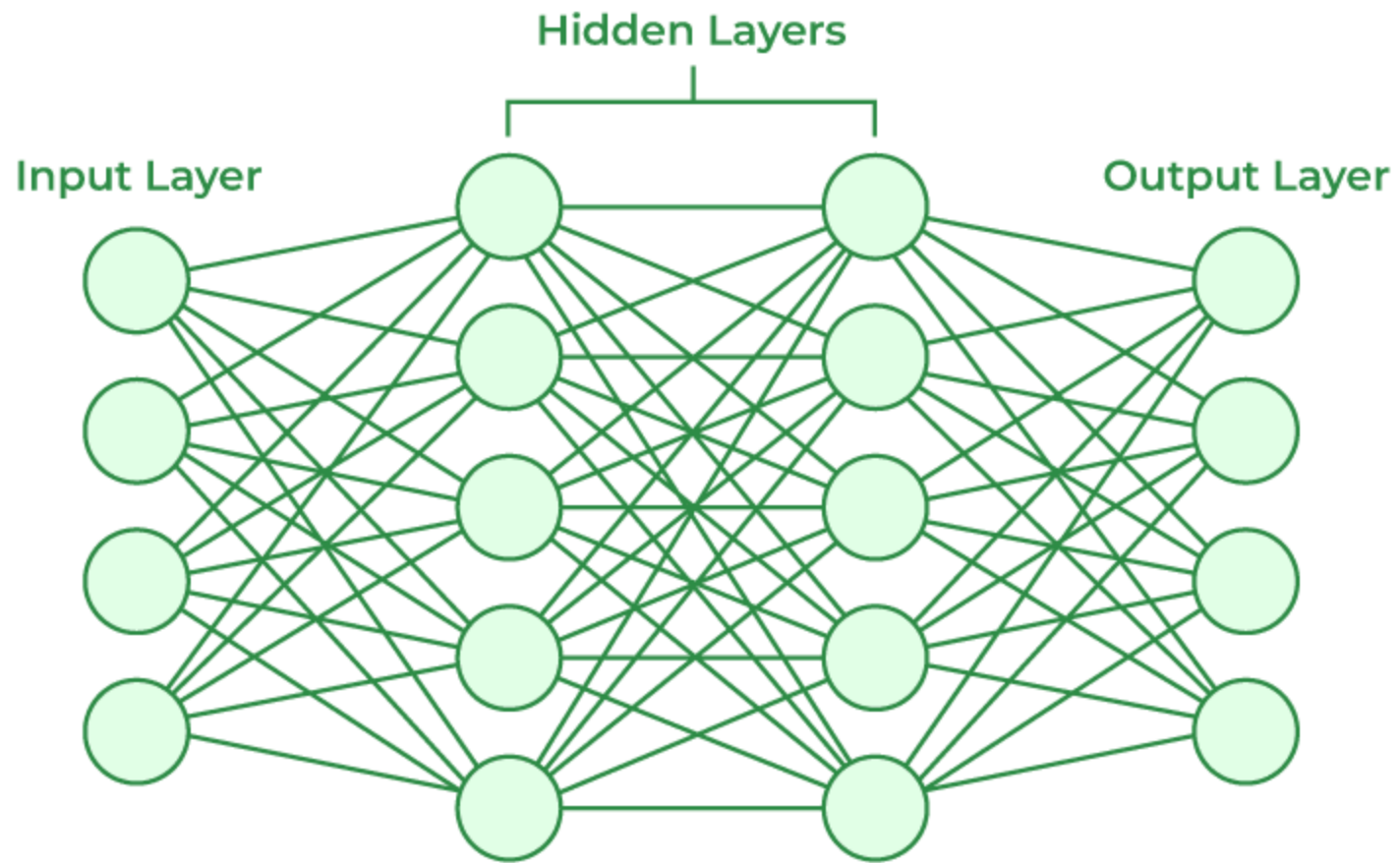
Information Technologies for Industrial Engineers

เทคโนโลยีสารสนเทศสำหรับวิศวกรอุตสาหกรรม

AI-powered Application

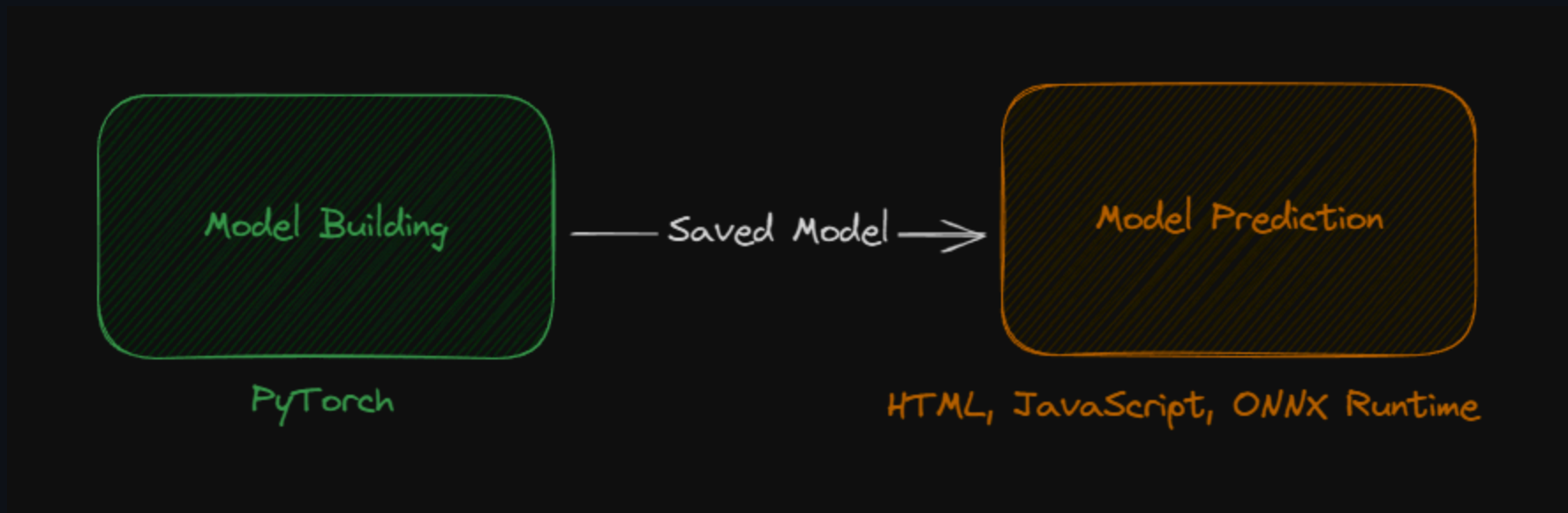
Artificial Neural Network

- A computing system modeled after the human brain's structure and function
 - Used in machine learning and AI to identify patterns, make predictions, and learn from data.
- Consist of interconnected **nodes** (artificial neurons) organized in **layers** (*input, hidden, and output*)
 - Information is processed through weighted connections and **activation functions**.



House Price Prediction (with ANN)

Model building



ONNX

- *Open Neural Network Exchange* ([Link](#))
- An open-source, standardized format for representing and exchanging machine learning models.
- Allows us to transfer model between different languages and runtimes.

Google Colab

- <https://colab.research.google.com/drive/1BdvM1OKY5lj6hmWHv9WP3FSImExgh9rr?usp=sharing>
- You should get `model.onnx`

Model prediction

| House Price Prediction (with ANN)

Setting up

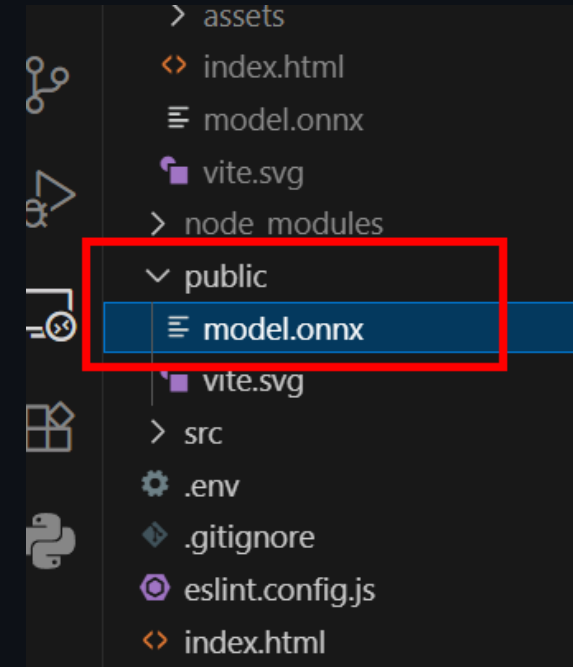
- `pnpm create vite@latest`
- ...

Library installation

- `pnpm install onnxruntime-web`
- `pnpm approve-builds`
 - Choose `protobufjs`

Model location

- Place the model in `./public/model` folder



vite.config.ts

```
import { defineConfig } from "vite";
import react from "@vitejs/plugin-react-swc";
export default defineConfig({
  plugins: [react()],
  assetsInclude: ["**/*.onnx"],
  optimizeDeps: {
    exclude: ["onnxruntime-web"],
  },
});
```

./src/model.ts

```
import * as ort from "onnxruntime-web";
export async function load_model() {
  // Path must be relative to public folder in Vite projects
  const modelUrl = `/model.onnx`;

  // Create session
  const session = await ort.InferenceSession.create(modelUrl);
  return session;
}
export { ort };
```

App.tsx

<https://github.com/it-for-ie-68/ml-house-price/blob/main/src/App.tsx>

Verify this

- Make sure that these values are correct!

```
function handleOutput(output: any) {  
  // Change here  
  const X_min = 800000;  
  const X_max = 2000000;  
  //  
}
```

```
function handleArea(input: string) {  
  // Change here  
  const mean = 1275.25;  
  const std = 353.78480394;  
  //  
}
```


CSS (Optional)

- Clear `import './index.css'` in `main.tsx`
- Insert this in `index.html`
 - [picocss](#)

```
<head>
...
  <link
    rel="stylesheet"
    href="https://cdn.jsdelivr.net/npm/@picocss/pico@2/css/pico.violet.min.css"
  />
</head>

</html>
```

Build and Deploy

- `pnpm run build`
- Deploy to [Netlify](#).